Clinical Cues as Spiking Neurons

A small, explainable seizure detector built from clinical EEG cues

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(Full technical slides and code private, available upon request)

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Seizure detection should be accurate and explainable

- EEG monitoring generates hours of data per patient, which is far more than experts can read in full
- A useful automatic seizure detector has to keep a delicate balance between catching seizures and minimizing false alarms
- Most recent detectors are large black-box networks which cannot explain an alarm
  - For example the SeizureTransformer won the 2025 seizure detection challenge and has 41 million parameters

The idea

- EEG data is turned into spiking neural events via rate, delta-sigma, and temporal encoding
- Small circuits of spiking neurons each watch for a clinical cue that a neurologist looks for
- Two small learned readouts (866 weights in total) combine the cues into a seizure probability every second, becoming an alarm
- Results: On the test set the model caught 72% of seizures at 1 false alarm per hour vs. 58% for SeizureTransformer



Clinical Cues for seizure detection

- New rhythm appears, clearly different from the background
- Keeps steady time cycle after cycle at a similar pace
- Frequency evolves usually slowing as the seizure progresses
- Sharp waves, spikes, or spike-and-wave (3/sec in absence seizures)
- Quiet onset where EEG flattens or low-voltage fast activity appears
- Starts in one place or on one side of the head
- Spreading from electrode to electrode
- Starts suddenly
- Not an artifact (less of a cue, more something to watch out for)

The base neuron model

A spiking neuron can be conceptualized as a leaky bucket, which fires when the bucket reaches a certain threshold.

$$v(t) = \beta\, v(t-1) + I(t), \qquad \beta = e^{-\Delta t/\tau}$$

v = the neuron's membrane potential

β = how much charge is kept each step, the leak

τ = how long it remembers

I = input arriving at this step

When v reaches a threshold θ, the neuron spikes and v drops back down.

Wiring this differently lets it detect different patterns!

Data

For this project, the Temple University Hospital Seizure EEG Corpus was used. This is one of the largest seizure datasets and contains ~1,500 hours of hospital EEG recordings.

The dataset is split into three subsections:

Dataset	Use	Recordings	Hours	Seizures
TUSZ v2.0.6 Train	Calibrate neurons, fit the heads	5,507	910	1,812
TUSZ v2.0.6 Dev	Choose alarm settings	2,203	436	492
TUSZ v2.0.6 Eval	Final test	880	129	360
Siena (Italy)	External test only	41	141	47

The Siena dataset was used to evaluate the model's ability to generalize to data it's never seen before.

Preprocessing

- Montage: 22 bipolar channels
  - Each = the difference between two neighboring electrodes, which cancels shared noise
- Resample to 250Hz to keep things consistent over recordings
- Filter 1-70Hz, plus a notch at the mains frequency (60Hz US, 50Hz Europe)
- Rescale each channel with a robust z-score to have all channels on the same scale

$$z = \frac{x - median(x)}{IQR(x)/1.349}$$

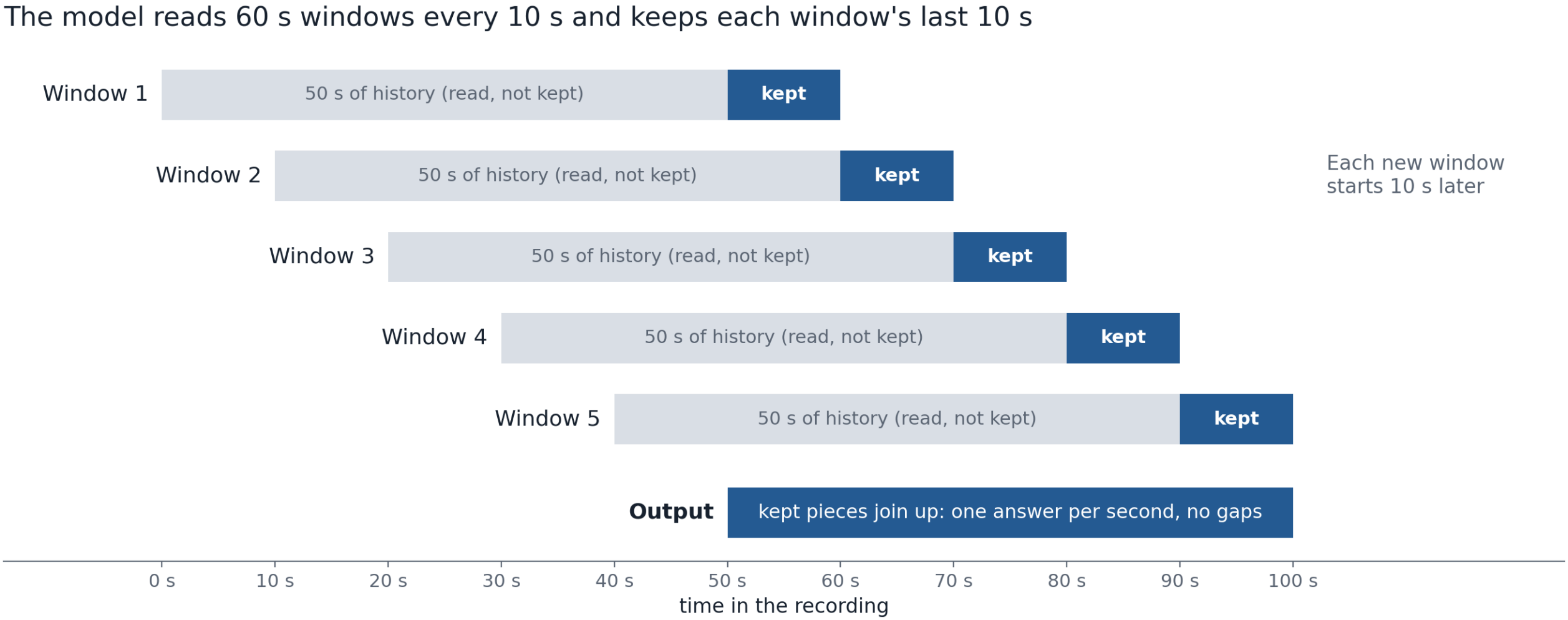
- Mask flat/dead channel so the model ignores them

Model gives one answer per second

- Each second is labelled a seizure if at least half of it falls inside an expert-marked seizure
- The model reads 60 seconds of EEG at a time, moving forward 10 seconds each step, and keeps the last 10 seconds of each window. Meaning every second is judged with 50 seconds of history
  - This window is for the neuron bank, some neurons need tens of seconds of EEG to build up.
- EEG samples every 4ms, and neurons update every 0.1 seconds (a tick), with one answer per second

Only the last step learns

1. EEG: 22 channels, 250 samples per second
2. Encoders: turn the EEG into spike trains
3. Clinical-cue neurons: 24 groups of fixed neurons, 72 outputs per channel, not trained
4. Summaries: Take each outputs mean (how widespread) and max (how strong somewhere) over channels, every second.
   So, 72 outputs per channel → mean & max over channel → 144/second
5. Two heads: small logistic regressions on
   1. Short: This second, 145 weights
   2. Long: this second plus its averages over the past 5, 15, 30, and 60 seconds, 721 weights
6. Alarm: smoothing plus threshold


The model reads 60 s windows every 10 s and keeps each window's last 10 s
Window 1
50 s of history (read, not kept)
kept
Window 2
50 s of history (read, not kept)
kept
Window 3
50 s of history (read, not kept)
kept
Window 4
50 s of history (read, not kept)
kept
Window 5
50 s of history (read, not kept)
kept
Each new window starts 10 s later
Output
kept pieces join up: one answer per second, no gaps
0 s
10 s
20 s
30 s
40 s
50 s
60 s
70 s
80 s
90 s
100 s
time in the recording

Three encoders turn EEG → Spikes

- Rate Encoder
  - Fires faster when a frequency band grows above the patient's baseline
  - Keeps a slowly updated running average of each band's size, divides the current size by it, and passes that ratio into a sigmoid to set a firing rate
  - Tracks a rhythm appearing relative to each patient's own background
- Delta-sigma Encoder
  - Fires a spike each time signal moves up or down by one step
  - Signal is divided by step size and rounded down to count how many steps higher/lower it is, steps up are ON spikes, steps down are OFF spikes
  - Tracks how fast and how far the signal moves, so steep edges give bursts of spikes

Three encoders turn EEG → Spikes, cont.

- Temporal Encoder
  - Marks the same point of every wave
  - Each wave gets one spike always at the same moment (when the wave rises through the middle line, zero), only waves at least 1.5x their usual size get a spike
  - Tracks when each wave happens so timing can be compared across channels

Cue 1: New rhythm

- Rhythm ALIF (adaptive integrate-and-fire)
  - Flags a rhythm that starts or changes
  - Its charge leaks a little each tick, and rate encoder's spike add to it, when it reaches the threshold it fires
  - Each spike also raises the threshold (fatigue), which then slowly relaxes so steady input stops firing it
  - Tracks the moment a rhythm begins

Cue 1: New rhythm, cont.

- Resonators
  - Each one rings at its own frequency
  - The state is represented as a complex number, think of it like pushing a swing. Each resonator is its own internal wave, swinging back and forth at its frequency and slowly dying down, and every incoming spike is a push
  - If a push (spike) arrives exactly one full cycle after the last, the swing has come all the way back to where it was, which is best moment to push, resulting in a kick up and the wave grows
  - When a push (spike) arrives half a cycle off, the swing is on the far side, which is the worst moment to push, resulting in a kick down and the wave shrinks
  - When a rhythm at the resonator's own frequency keeps pushing in time, the wave builds up until it fires
  - Tracks a regular rhythm and its frequency even when it's small

Cue 2: Rhythm keeps steady time

- Regularity neuron
  - Checks that a rhythm keeps time
  - It measures each cycle's length (time between upward zero crossings) and compares with previous one
  - A cycle counts as regular if the two agree within a small tolerance, a running average gives the share of recent cycles that kept time, no size term
  - Tracks steady rhythms, including seizures no bigger than background
- Persistence neuron
  - Measures how long one rhythm has dominated
  - Each tick asks 'is one resonator firing strongly right now?', and keeps a slow running average of the answer, it never tires so it keeps rising in long seizures
  - Tracks the sustained part of a seizure after onset neurons fade

Cue 3: Evolves in Frequency

- Frequency drift neuron
  - Fires when a rhythm hands off from one frequency to the next
  - Leaky sum adds the faster resonators output from a moment ago (delay line) to the slower neighboring resonators output now
  - Threshold is set so one input alone can’t reach, only both together, fast then slow, can cross it
  - Mirror version detects speeding up (slow then fast)
- Slow drift neuron
  - The same over a longer delay
  - Uses smaller of the two inputs as an AND so both rhythms have to be there
- Tracks the classic slowing of seizures, steady artifact rhythms never evolve so they never trigger it

Cue 4: Sharp waves

- Sharpness neuron
  - Detects steep edges
  - Leaky sum of ON or OFF spikes with very short memory, only a burst arriving within a few ms can reach threshold
  - Tracks spikes and sharp waves
- Spike-and-wave neuron
  - Detects the absence-seizure pattern
  - Pair is a sharp spike followed shortly by a slow-wave resonator, decaying counter adds each pair and fires when several arrive in a row
  - Tracks 3-per-second spike-and-wave characteristic

Cue 4: Sharp waves, cont.

- Rhythmic-sharp neuron
  - Detects sharp waves at steady pace
  - It times the gap between one sharp spike and the next, a gap counts if it’s a certain length and about the same as the gap before it, i.e. the sharp waves are keeping a steady beat
  - Each matching gap adds a point, and two points close together make it fire
  - Tracks periodic discharges

Cue 5: Quiet Onsets

- Silence neuron
  - Fires when the EEG suddenly goes flat
  - Constant upward drive scaled to the channel's normal activity is pushed down by current activity, i.e. normal EEG keeps it pinned down, when activity stops, the drive lifts it to threshold
  - Tracks flattening at onset
- Fast activity neuron
  - Catches small, fast rhythms
  - The same swing-and-push resonator as in Cue 1 tuned to higher frequency
  - Low-voltage fast activity is small so the slow waves are filtered out (>14Hz), uses smaller steps, and fires at a lower threshold
  - Tracks low-voltage fast activity

Cue 5: Quiet Onsets, cont.

- Line length neuron
  - Detects activity getting busier, not bigger
  - Measures how much the signal moves up and down for its size, then compares that with the last 20 seconds
  - Below 0 = smoother, 0 = normal, 1 = twice as jagged
  - Tracks changes in texture when amplitude stays the same

Cue 6: Starts in one place

- Hemisphere comparator
  - Fires when one side is steadily busier
  - Leaky sum adds this channel's evidence minus that of its mirror channel on the other side
  - Equal activity cancels, a steady imbalance builds up to threshold
  - Tracks one-sided onsets
- Local contrast + top-3
  - Finds a channel that stands out
  - Each channel's rhythm minus the average of its montage neighbors, the average of the three strongest gives one 'how focal number'
  - Tracks a local hot spot

Cue 7: Spreads

- Spread neurons
  - Detect activity travelling from one electrode to the next
  - Each channel's spikes leave a short fading trace and the neurons adds a neighbors trace delayed by a fixed amount to this channel's current trace only firing if both line up
  - Several delays act as several speed sensors
  - Tracks spread and speed

Cue 8: Sudden Onset

- Onset neuron
  - Fires when several cue groups jump together
  - Each of six detector groups is compared with its own recent average in units of its normal variability
  - A group jumps if the change is larger than usual, and when enough groups jump at once, the potential rises
  - Tracks an abrupt break from background

Cue 9: Not an artifact

- Synchrony + artifact signatures
  - Recognizes movement and bad electrodes
  - Synchrony sums short spike traces across all channels and fires when many coincide
  - Signatures check for things that the brain can’t produce (clipping, flat line, step jumps, muscle noise)
  - Artifact evidence needs both
- Artifact gate
  - Decides whether to trust rhythm neurons
  - An artifact neuron and a rhythm neuron inhibit each other (each spike pushes the rival down)
  - The gate closes only when the artifact neuron has been winning more recently than a rhythm neuron
- Tracks artifacts without throwing away real seizures

Two heads: This second, and the past minute

- Every second each of the 72 neuron outputs is summarizes as its mean and max across channels: 144 numbers
- Short head
  - A weighted sum of this second's 144 numbers, turned into a probability by sigmoid
- Long head
  - The same, plus the averages of those 144 numbers over the past 5, 15, 30, and 60 seconds
- Final score is the larger of the two

Turning scores into alarms

1. Smooth the score with a 5-second moving average
2. Start an alarm when it reaches the threshold, keep it going while it stays above 0.75 * threshold
3. Merge alarms less than 3 seconds apart, no minimum alarm length so short seizures survive

Metrics & SeizureTransformer Context

Metric	What it means
Sensitivity	Share of seizures caught (an alarm overlaps the seizure)
False alarms per hour (or per 24hr)	Alarms that touch no seizure/hrs of recording
Precision	Share of alarms that are real seizures
AUROC	Chance that a random seizure second scores higher than a random normal second (1.0 = perfect, 0.5 = guessing)

SeizureTransformer (Wu et al. 2025): Winner of the 2025 seizure-detection challenge. Run with its released weights on the same TUSZ eval set, scored with same smoothing, threshold choice on dev.

Results Comparison

	This model	SeizureTransformer
Seizures caught at 1 FA/hr	71.9%	57.5%
Seizures caught at 0.68 FA/hr	65.8%	54.7%
Per-second AUROC	0.925	0.893
60 second window AUROC	0.933	0.888

For each model the alarm threshold is slid until it produces exactly 1 or 0.68 per hour on the eval dataset.

Settings chosen on dev (1 FA/h)	This model	SeizureTransformer
Seizures caught	58.9%	53.1%
False alarms per hour	0.46 = 11 FA/day	0.77 = 18 FA/day
Precision	78.7%	67.3%
F1	0.674	0.593
Median delay after onset	4.1 sec	5.0 sec

These settings were fixed on dev before testing, so this is what the model would do if deployed as-is

Specific Seizure Type Results

Seizure Type	Caught	Seizures
Tonic-Clonic	100%	8
Generalized	76%	108
Complex partial	68%	41
Focal	47%	153
Absence	45%	49

Note: This is on the 1FA/hr setting and, the eval group is small so it's noisy, one patient can move these numbers quite a bit

Long seizures were detected easier; short seizures and slow onsets were harder.

Duration	Caught	Seizures
Under 10 sec	41%	54
10-30 sec	23%	86
30-90 sec	77%	149
Over 90 sec	79%	71

Freq. @ onset	Caught	Seizures
Delta (<4Hz)	28%	71
Theta (4-8Hz)	63%	186
Alpha (8-13Hz)	74%	82
Beta (13Hz≥)	62%	21

Siena evaluation

	TUSZ eval	Siena (Italy)
Seizures caught (settings from TUSZ dev)	58.9% at 0.46 FA/hr = 11/day	61.7% at 0.66 FA/hr = 16 FA/day
Seizures caught at 1 FA/hr	71.9%	68.1%
Per-second AUROC	0.925	0.870
Precision	78.7%	25.0%
Median delay	4.1 sec	15 sec

- Nothing was retuned for this dataset
- Precision is lower because Siena has far fewer seizures per hour of recording so each false alarm weighs more
- Delays are longer because Siena's experts mark onsets earlier

Iterations

- The first full clinical-cue model reached AUROC of 0.85 and caught 43.7% of seizures
- Then the loop was: find seizures that were missed → trace why → add a targeted fix → re-test on unseen patients
- Final model reaches and AUROC of 0.925 and caught 72% of seizures

Failure	Cause	Fix
Absence seizures missed	Artifact filter mistook synchronized spike-and-wave for movement	Spike-and-wave neurons; let an established seizure rhythm override filter
Slow seizures no bigger than background (invisible)	Detectors mostly measured size	Regularity neuron; steady timing whatever the size
Long seizures faded out	Neurons adapted to them like background	Persistence neuron, which doesn't tire
Short seizures erased	Too much smoothing in alarm rule	Lighter smoothing, no minimum alarm length
Context helped long seizures but hurt short ones	One readout couldn't do both	Two readouts (short and long), take the more confident one

Two heads

	Short head only (145)	Long head only (721)	Both, max(866)
All seizures	67.2%	65.3%	71.9%
Absence seizures	69.4%	8.2%	44.9%
Seizures over 90 sec	71.8%	78.9%	78.9%

- Context (long head) is what catches the long seizures, but alone it erases almost all absences
- Taking the max keeps most of both and gives best overall result, but absences still drop compared with the short head alone

Limitations

- Limitations
  - Focal seizures: 47% caught, focal-extent only 24%
  - Slow (delta) onsets: 28%
  - 10-30 sec seizures: 23%
  - Not yet real time as each recording is rescaled using statistics from whole recording
  - Many seizure types have under 50 examples and dominated by a few patients

Next Steps/Future Work

- Creating a better model: The cool part about this approach is that parts are modular, everything can be improved or replaced independently
  - Try using different encodings methods
  - Try using different neuron models
- Real time version by replacing whole-recording rescaling with running estimate
- More external dataset validation
- Run neuron bank on neuromorphic hardware and measure power

How it was built (AI Disclaimer)

AI assistance (Claude, Gemini) was used for the code and analysis of this project.

Supervised at the Center for Translational Cognitive Neuroscience by Principal Investigator Dr. Brandon Frank

Computing: Boston University Shared Computing Cluster